

Report of Grover's Algorithm Analysis

Who did what

This report details exactly who focused on which part of the explanation of the algorithm

Task	Did by :
Algorithm Theory (Purpose, Methods, Explanation of the algorithm)	Antoine
Practical Demonstration (Small-scale implementation, Code walkthrough, Verification of iteration formula)	Andrea
Critical Analysis (Limitations of the algorithm)	Andrea
Video Production (Recording and Editing)	Antoine
Report & Sources (Creation of this PDF)	Andrea

References

Here is a list of the references used for our presentation of Grover's Algorithm

1. *Quantum Computation and Quantum Information*, Nielsen, M. A., & Chuang, I. L.
<https://profmcruez.wordpress.com/wp-content/uploads/2017/08/quantum-computation-and-quantum-information-nielsen-chuang.pdf>
2. **QuTiP / Qiskit Documentation:** <https://qutip.readthedocs.io/en/qutip-4.7.x/>
3. **Grover's Original Paper (Optional but good):** *A fast quantum mechanical algorithm for database search*, L. K. Grover (1996).
4. IBM explanation : <https://quantum.cloud.ibm.com/docs/en/tutorials/grovers-algorithm>
5. But what is Quantum Computing ? (Grover's Algorithm) of 3Blue1Brown
<https://www.youtube.com/watch?v=RQWpF2Gb-gU&t=2016s>
6. Gemini 3 for helping with the Python Algorithm